# AssistVision: A Turntable–Load-Cell Rig and a Controlled Benchmark of Monocular versus Multi-View Food Volume Estimation

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***Abstract*—Image-based portion estimation needs metric volume, and a single image does not contain metric scale. We build a low-cost capture rig (ESP32-S3 camera, stepper turntable, HX711 load cell, printed ArUco mat) that returns $N$ calibrated views together with the true mass of the plate, and we ask, with controlled experiments rather than anecdotes, when a single-view depth network can replace multi-view reconstruction. On the 507-dish Nutrition5k depth test split we benchmark [n models] monocular depth models under three explicitly separated calibration protocols (raw, deployable anchor, oracle upper bound) and report bootstrap confidence intervals and paired tests. On [n synth] synthetic plates with exact mesh volume we compare dense multi-view stereo with monocular depth and ablate view count, sensor resolution, JPEG quality, marker-scale error and plate material. [abstract numbers]**

## I. Introduction

Dietary assessment by photograph is attractive because it removes manual logging, but portion size—grams—is the quantity nutrition depends on and the one images recover worst. Perspective projection discards absolute scale: a small plate near the camera and a large plate far away produce the same pixels. Any system must inject metric scale from outside the image: from a reference object, from a depth sensor, from camera motion with known geometry, or from a learned prior. This paper measures how much each of these choices costs.

Our contributions are:

1) An open hardware capture rig (Sec. IV) that produces multi-view images, metric scale and gravimetric ground truth in one button press, with firmware, host software, wiring and a bench-test protocol.
2) A benchmark of monocular depth models for food on real RGB-D data (Nutrition5k) that separates deployable calibration from oracle calibration, a distinction prior food-volume work often blurs.
3) A controlled synthetic study with exact volumes that isolates the factors the rig designer actually controls: views, resolution, compression, marker accuracy and serveware material.
4) A documented record of design decisions and rejected alternatives (Sec. III), each tied to a measurement or a stated assumption.

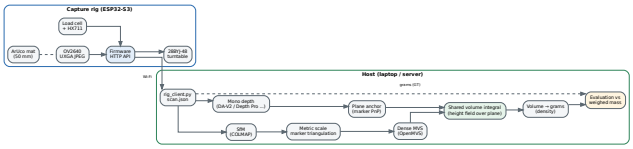



Fig. 1. System architecture. The rig exposes an HTTP API; the host drives the scan loop, so dropped frames are retried rather than corrupting a scan.

## II. Related Work

**Food volume.** Nutrition5k [1] provides overhead RGB-D video and weighed dishes; its depth-based volume baseline uses a constant per-pixel footprint at the capture-plane distance. Multi-view food reconstruction has used structure-from-motion with reference cards and dense stereo [2], [3]. **Monocular depth.** Relative depth models (MiDaS [4], Depth Anything [5], [6]) predict depth up to an unknown affine transform of inverse depth; metric models (ZoeDepth [7], Depth Pro [8], metric-finetuned Depth Anything V2) predict absolute depth from priors learned on room- and street-scale scenes. **Reconstruction.** We use COLMAP [9] for SfM and OpenMVS for dense MVS.

## III. Design Decisions and Rejected Alternatives

Table I summarises each decision; the evidence column points to the experiment that supports it. Decisions marked *assumption* rest on datasheets or cost quotes, not on our measurements.

## IV. Hardware

The rig (Fig. 1, 2) uses an ESP32-S3 camera board (OV2640, up to $1600 \times 1200$), a 28BYJ-48 stepper through a ULN2003 driver (4076 half-steps per revolution, 0.088° per step), and a 5 kg bar load cell read by an HX711 at 10 Hz. The firmware exposes `/capture`, `/rotate`, `/weight`, `/tare`, `/calibrate` and camera-setting endpoints. Exposure, gain and white balance are locked after convergence so that all views of a scan share photometry, which MVS assumes. Stepper coils are released while weighing to avoid heating and vibration of the load cell. Tare and the counts-per-gram

TABLE I
DESIGN DECISIONS, ALTERNATIVES CONSIDERED, AND THE EVIDENCE BEHIND EACH CHOICE.

| Decision | Chosen | Rejected alternatives | Why / evidence |
|---|---|---|---|
| Scale source | Printed ArUco mat, multi-view triangulation | Credit-card reference in one image; metric monocular depth; depth sensor | Volume scales with the *cube* of length error (Sec. V-B); metric networks are off by large factors at plate distance (Table **??**); a D435-class sensor costs $\sim$10$\times$ the camera (assumption). |
| Geometry | Dense MVS (COLMAP + OpenMVS) | Sparse SfM points only; single-view depth | Sparse points under-cover low-texture food (Sec. VI-C); single view cannot see the far side of a mound even with perfect depth. |
| Capture motion | Fixed camera, motorised turntable | Hand-held phone orbit; multi-camera ring | Repeatable baselines and known angles; one camera keeps cost and synchronisation trivial; the angle prior is checked against SfM. |
| Camera tilt | 45° elevation | Straight overhead (Nutrition5k) | Overhead view makes the support plane fronto-parallel: the relative-depth anchor fit becomes rank-deficient (condition number $10^{14}$, Sec. V-C). |
| Controller | ESP32-S3 + OV2640 | Raspberry Pi + camera; smartphone app | Pi adds cost, boot time and an OS to maintain; phones vary per user and cannot drive the turntable or scale. Processing is host-side, so the MCU only needs camera, Wi-Fi and GPIO (assumption: cost table in hardware README). |
| Ground truth | 5 kg bar load cell + HX711 under the plate | External kitchen scale; no mass at all | In-rig weighing removes a manual step and lets every scan carry its own reference; tare-before-serve gives food-only mass. |
| Volume integral | Height field over the fitted support plane | Constant per-pixel footprint (Nutrition5k); voxel carving; mesh + watertight closure | Exact on tilted planes in our unit tests (0.09 % error) where the constant-footprint formula is not; shared across all methods so comparisons measure geometry, not bookkeeping. |
| Mono calibration | Report raw / anchor / oracle separately | Report only oracle-aligned numbers | Oracle alignment uses the ground truth being evaluated; it is an upper bound, not a deployable result. |

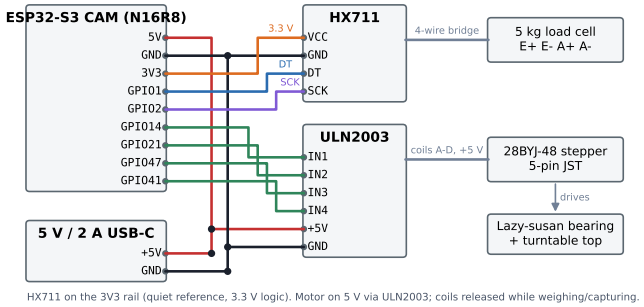



Fig. 2. Wiring of the ESP32-S3, HX711 load-cell amplifier and ULN2003 stepper driver.

calibration persist in flash. The firmware compiles for the ESP32-S3 target with the Arduino core 3.3.12; a fake-firmware test exercises every client endpoint, retry on a dropped frame, and calibration arithmetic.

**Physical validation (pending).** Load-cell linearity, repeatability and drift, turntable angular error, scan duration and transfer latency are specified as scripted bench tests (`rig_client.py bench-*`) whose results are [to be measured on the assembled rig]. We report no physical number we have not measured.

## V. METHOD

### A. Shared volume integral

All methods produce a 3-D point set $P$ in metres and a support plane $(\mathbf{n}, d)$. Points are projected into an orthonormal plane basis, binned into square cells, and each cell keeps the median (dense MVS) or the back-projected height $h = \mathbf{n}^\top \mathbf{p} + d$. Empty cells inside the footprint are filled by nearest neighbour, negative heights are clipped, and $V = \sum_{\text{cells}} h\, c^2$. Because every method shares this integral, differences in volume reflect differences in geometry.

### B. Metric scale from the marker mat

Marker corners are detected in every view and triangulated with the SfM cameras; the scale $s$ is the ratio of the known side to the median triangulated side. A relative error $\epsilon$ in $s$ produces a volume error of $(1+\epsilon)^3 - 1 \approx 3\epsilon$, so a 1 % scale error costs about 3 % in volume; Sec. VI-C measures this directly.

### C. Monocular protocols

**Raw**: the model's metric output is used as-is. **Anchor**: the prediction is calibrated using only the support plane (the table

in Nutrition5k, the marker mat in our rig), never the food; metric outputs get one scale, relative outputs an affine map of inverse depth. **Oracle**: the same maps fitted to ground-truth depth over the whole dish region—an upper bound that is not available at deployment. For relative models we record the condition number of the anchor fit; a fronto-parallel plane has nearly constant inverse depth, so scale and shift are not separately identifiable.

## VI. Experiments

### A. E1: Monocular depth on Nutrition5k

[e1 results]

### B. E2: Volume versus weighed mass

[e2 results]

### C. E3: Multi-view versus monocular on exact synthetic volumes

[e3 results]

### D. E4: Ablations

[e4 results]

## VII. Limitations

[limitations results]

## VIII. Conclusion

[conclusion results]

## Reproducibility

All code, scene specifications, per-dish result files and the scripts that generate every table and figure are in the repository. Evidence categories are labelled throughout: *Nutrition5k* (public real RGB-D data), *synthetic* (Blender/Cycles renders with exact mesh volume), *VM* (timings on an 8-core CPU-only cloud VM, not on the ESP32), and *physical* (pending rig measurements).

## References


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